

A prospective study of prepregnancy dietary fat intake and risk of gestational diabetes

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Background: Fatty acids play a vital role in glucose homeostasis; however, studies on habitual dietary fat intakes and gestational diabetes mellitus (GDM) risk are limited and provide conflicting findings. Objective: We determined whether the total amount and the type and source of prepregnancy dietary fats are related to risk of GDM. Design: A prospective study was conducted in 13,475 women who reported a singleton pregnancy between 1991 and 2001 in the Nurses' Health Study II. In these women, 860 incident GDM cases were reported. The adjusted RR of GDM was estimated for quintiles of total fat, specific fat, and the source of fat intakes by pooled logistic regression. Results: Higher animal fat and cholesterol intakes were significantly associated with increased GDM risk. Across increasing quintiles of animal fat, RRs (95% CIs) for GDM were 1.00 (reference), 1.55 (1.20, 1.98), 1.43 (1.09, 1.88), 1.40 (1.04, 1.89), and 1.88 (1.36, 2.60) (P-trend = 0.05). Corresponding RRs (95% CIs) for dietary cholesterol were 1.00 (reference), 1.08 (0.84, 1.32), 1.02 (0.78, 1.29), 1.20 (0.93, 1.55), and 1.45 (1.11, 1.89) (P-trend = 0.04). The substitution of 5% of energy from animal fat for an equal percentage of energy from carbohydrates was associated with significantly increased risk of GDM [RR (95% CI): 1.13 (1.08, 1.18); $P < 0.0001$]. No significant associations were observed between dietary polyunsaturated fat, monounsaturated fat, or trans fat intakes and GDM risk. Conclusion: Higher prepregnancy intakes of animal fat and cholesterol were associated with elevated GDM risk.